Multimodal Pedagogical Friction Detector

Fusing Facial Affect · Talk Moves · LLM Reasoning to Surface Teachable Moments at Scale

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Motivation

Why this problem?

Manual classroom coding takes years of expert effort and cannot scale to real-time teacher feedback [1].

The challenge

- When students are confused, the teacher's **next move** is critical for deep learning [2]
- Existing tools are **single-modal**: transcript *or* affect — never fused with semantic reasoning
- Edu-ConvoKit & TutorCoPilot ignore student affect; affect systems ignore teacher response quality

Our goal

- Define **Pedagogical Friction**: elevated student confusion **and** a low-quality teacher move in the same 30-second window
- Fuse **vision** (DeepFace) + **language** (ELECTRA + `claude-sonnet-4-6`)
- Scale to **53 TIMSS lessons, 7 countries** without human labelling

Problem Setup

Definition of Pedagogical Friction

A 30-second window where student confusion is *elevated* **AND** the teacher's talk move is *low-quality* (answer-giving, closed loops, transmission-only instruction).

Video data

- TIMSS AU1 lesson (10 min)
- 1 200 frames → DeepFace @ 2 Hz
- 87 bins | 1 human positive

Transcript corpus

- 53 TIMSS lessons
- 28 math + 25 science
- 7 countries | 5 113 bins

Annotation

- Human: 1 positive (bin 11)
- Heuristic: 31.5% flagged
- ELECTRA: 19.3% flagged

Shared 30-second temporal grid aligns vision & language streams for every lesson.

Slide 3 / 5 Pipeline Architecture

Video
DeepFace 2 Hz
confusion c_t

→

Audio/Text
Faster-Whisper
→ ELECTRA

→

30-s Grid
$\bar{c}_k$ per bin
z-score filter

→

Claude
sonnet-4-6
friction verdict

→

Output
`friction: bool`
alt. strategy

Confusion proxy (Eq. 1)

$$c_t = \sum_{e \in \mathcal{E}} w_e \cdot p_t(e)$$

$w_{\text{fear}}=0.90,\ w_{\text{sad}}=0.85,\ w_{\text{disgust}}=0.50\ldots$

Candidate filter (Eq. 2)

$$\bar{c}_k > \mu_C + z\,\sigma_C \ \wedge\ q_k \neq \mathit{high}$$

$z=1.0$ flags the top $\sim$16% of bins.

Talk Move fallback: **(1)** ELECTRA → **(2)** Edu-ConvoKit → **(3)** Keyword heuristic

Experiments

Track 1: AU1 Full Video

- 1 200 frames → DeepFace + Whisper + Claude
- 7 candidates → Claude rejects **all** (social disruption)
- Wide-angle: $\bar{c}_k = 0$ every bin — **vision fails**

Track 2: 54-Lesson Scale-Up

- SLURM GPU array (NVIDIA A40)
- 5 318 bins | 1 797 flagged
- Claude confirms **21** moments (0.4%)
- Wide-angle failure confirmed

Track 3: Human Eval (87 bins)

- 1 annotator → **1 positive** (bin 11)
- Heuristic: F1 = 0.10
- ELECTRA: F1 = 0.00
- Heuristic + Claude: **F1 = 1.00**

Key finding

$r = -0.35$ correlation between heuristic & ELECTRA rates across 53 lessons ⇒ **annotation methodology is a first-class design variable**.

Results & Takeaways

Condition	Bins	Flag	%	F1
Human evaluation (AU1, 87 bins)				
Heuristic	87	20	23%	0.10
ELECTRA	87	33	38%	0.00
+ Claude	87	1	1%	**1.00**
Transcript-only (53 lessons)				
Heuristic	5113	1611	31.5%	—
ELECTRA	5113	988	19.3%	—
+ Claude	5113	211	4.1%	—

- **F1 = 1.00**
  Claude cuts false positives 20×; zero missed detections
- **7.6× reduction**
  1 611 flags → 211 confirmed friction moments
- $r = -0.35$
  Heuristic vs ELECTRA anti-correlated across 7 countries
- **Vision limit**
  Wide-angle ⇒ DeepFace unusable; language is the primary signal

References I

Washington, DC: National Center for Education Statistics, 2003.